VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi-590 018

A Mini - Project Report

On

"BIBLIOTRACK - AN AI POWERED BOOKSTORE"

Submitted in partial fulfillment of the requirements for the **MINI PROJECT (BCD586)**

course of the 5th semester

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In

Computer Science & Engineering (DATA SCIENCE)

Submitted by

Ms. Harshitha S (4AI23CD019)

Ms. Prajan K S (4AI23CD038)

Ms. Ramya Manjunath Gouda (4AI23CD041)

Ms. Shalini M B (4AI23CD048)

Under the guidance of

Ms. GAGANA DEEPA J **B.E., M.Tech.**

Assistant Professor

Department of CS&E (DATA SCIENCE)

Adichunchanagiri Institute of Technology

CHIKKAMAGALURU - 577102

2025-26

ADHICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY

Jyothinagar, Chikkamagaluru-577102

DEPARTMENT OF CS&E (DATA SCIENCE)

CERTIFICATE

This is to certify that the Mini project work entitled **"BIBLIOTRACK-AN AI POWERED BOOKSTORE"** is a bonafied work carried out by **Ms. Harshitha S (4AI23CD019), Ms. Prajna K S(4AI23CD038), Ms. Ramya Manjunath Gouda (4AI23CD041),Ms. Shalini M B(4AI23CD048)** in partial fulfillment for the **Mini Project (BCS586)** course of 5th semester Bachelor of Engineering in **Computer Science and Engineering (Data Science)** of the Visvesvaraya Technological University, Belagavi during the academic year **2025-2026**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for the said Degree.

Signature of the Guide	**Signature of Coordinator**
Ms. Gagana Deepa J B.E., M.Tech	**Mrs. Shilpa K V.** B.E., M.Tech
Assistant Professor	**Assistant Professor**

Signature of the HOD
Dr. Adarsh M J B.E., M.Tech., Ph.D
Associate Professor and Head

ABSTRACT

Book discovery and intelligent book management have become essential components of modern digital reading platforms. With the rapid growth of online bookstores, readers often struggle to find books that match their interests due to limited search capabilities, lack of personalization, and absence of interactive assistance. Traditional online systems that rely solely on basic keyword searches or static filters are time-consuming, inefficient, and often fail to provide users with meaningful recommendations. Bibliotrack, an AI-powered bookstore system, is developed to overcome these limitations by offering intelligent, automated, and user-friendly book discovery functionalities.

This project implements a robust and interactive online book discovery and management system that enhances user engagement through features such as personalized recommendations, AI chatbot assistance, visual book search, and community book clubs. The platform provides dedicated interfaces for customers and administrators, supporting seamless book browsing, smart search, book club participation, and backend inventory management. Bibliotrack is built using a combination of modern technologies including Django for backend development, HTML/CSS/JavaScript for the frontend, SQLite for database management, and AI modules such as CNN-based visual search and NLP-based chatbot interaction. These technologies collectively ensure high performance, accuracy, and intelligent automation across all modules of the system.

Overall, this project delivers a reliable, efficient, and AI-driven approach to online book discovery and management. By integrating smart recommendation engines, visual identification models, and conversational chatbot features, Bibliotrack significantly enhances the reading experience while enabling administrators to manage books effectively. The system contributes to improved user satisfaction, faster book discovery, and a modernized digital bookstore ecosystem suitable for real-world deployment.

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Harshitha S

Prajana K S

Ramya Manjunath Gouda

Shalini M B

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Chapter 1

INTRODUCTION

1.1 Background

- **Context**: Books play a significant role in learning, personal growth, and entertainment. However, with the rapid expansion of digital bookstores, users often find it difficult to efficiently discover books that match their interests. Traditional online bookstores only provide basic search and purchase features, lacking intelligent assistance that modern readers expect. As digital reading increases, users require smarter, personalized, and more interactive systems for book discovery.
- **Problem**: Existing online bookstores depend mainly on basic text search and limited filters. Users often struggle to find the right books due to the absence of personalized recommendations, lack of visual search, no AI assistant for guidance, and no community space. This makes book discovery slow and ineffective, especially for new readers.
- **Opportunity**: Advancements in AI, ML, and NLP enable a smarter bookstore system. BiblioTrack integrates AI-powered recommendations, visual search through book c overs, an interactive chat assistant, and a community book club. These features create a faster, more intuitive, and engaging book discovery experience for all users.

1.2 Problem Statement

- **Overview of the Problem**: Most existing bookstore platforms rely heavily on manual text-based search and limited filtering options.
- **Specific Issues**:
  - Lack of personalized recommendations
  - Inability to visually identify books (e.g., by cover images)
  - Absence of an interactive assistant for guidance
  - No community space to connect with fellow readers

1.3 Objective of the System

- To develop an AI-powered book discovery system that offers personalized book recommendations based on user preferences, reading history, genres, and interests.
- **Key Goals**:
    - **Provide accurate and unbiased book recommendations** tailored to each user.
    - **Support AI-powered chat assistance** to help users with queries and book suggestions
    - **Enable visual search** to identify books using cover images
    - **Suggest related or alternative books** when a preferred title is unavailable
    - **Allow admins to easily manage books, categories, and listings** through a dedicated dashboard
    - **Offer smart search and category-based discovery** for faster results

1.4 Significance of the System

- **Fair Access:** Ensures all readers, including beginners, get equal access to quality book suggestions
- **Personalization:** Recommends books based on each user's reading history, interests, and preferences
- **Inclusivity**: AI chatbot assists users with easy, conversational guidance for book discovery
- **Digital Empowerment:** Helps users confidently explore new genres, authors, and book categories
- **Data-Driven:** Book suggestions are powered by AI, ML, and intelligent recommendation algorithms
- **Reading Awareness**: Suggests similar or related books to help users expand their knowledge and interests

1.5 Scope of the Project

In Scope:

- AI-based book recommendation engine
- Visual Search for Book Identification
- AI Chatbot Assistance
- Admin Dashboard for Book & Category Management
- Bookclub to interact with other peers

Out of Scope:

- Biometric authentication or identity verification
- Guaranteed book availability or delivery services
- Offline reading or physical library management modules

1.6 Methodology

- The system uses a web-based frontend (HTML, CSS, JavaScript),backend(Django)
- AI modules (TensorFlow.js) to manage books and generate intelligent recommend
- NLP analyzes user inputs, genres, and reading preferences to improve book suggestions.
- AI models identify books and compute similarity scores for recommendations through visual search.
- The chatbot uses AI-driven text interaction to guide users, answer queries, and assist in book discovery.
- Agile methodology ensures iterative improvements in recommendations, search accuracy, and overall user experience.

1.7 Target Audience

- **Readers/Users**: Individuals looking for personalized book recommendations smart search features
- **Admins**: Managers responsible for adding, updating, and organizing books, authors, and categories

- **Community Members**: Users participating in the Book Club for discussions and sharing reading interests

1.8 Overview of the Report

This report covers system design, implementation, testing, results, and future enhancements for the **Online Bookstore System.** The following chapters include:

- **Chapter 1:Introduction** – Describes the brief introduction and design of the system.
- **Chapter 2: System Design** – Describes the architecture and design of the system.
- **Chapter 3: Implementation** – Discusses the system's development and the technologies used.
- **Chapter 4: Testing and Validation** – Details the testing process and results.
- **Chapter 5: Results and Discussions** – Presents and results obtained and discusses the limitations
- **Chapter 6: Conclusion and Future enhancement -** Summarizes the project and suggests future improvements.

Chapter 2

SYSTEM DESIGN

This chapter describes the technical design of the AI-powered book discovery and recommendation engine for Bibliotrack, explaining its architecture, components, and how they work together to enable intelligent book search, visual identification, and personalized recommendations.

2.1 System Architecture

Bibliotrack follows a client–server architecture **integrated with an AI layer.**

Client– Server Model:

- Users (students, readers, admins) interact with the web UI.
- The backend processes requests, manages book data, and communicates with AI services.

AI Layer: Handles all intelligent tasks:

- **Visual Search Module**: Identifies books from images using CNN + Image Hashing.
- **Recommendation Engine**: Suggests similar or trending books using ML models.
- **Chatbot Module**: Provides reading suggestions and answers queries using NLP.

Components:

2.1.1 Frontend (User Interface)

- Book browsing dashboard
- Visual search/upload interface
- User profile & wishlists
- Book club/community pages
- Admin dashboard for inventory & announcements

2.1.2 Backend Server

- Handles authentication
- Book search logic
- Image upload handling

- Recommendation & similarity computation
- Book club post management
- Order/inventory operations

2.1.3 Database

Stores :

- Users
- Books (title, genre, description, cover embeddings)
- Book categories & authors
- Book club posts & comments
- User history (search, reading lists)
- AI logs (embeddings, image hashes)

2.1.4 AI Engine

- CNN model for visual search
- Embedding generator (CNN)
- Recommendation model using similarity (KNN/embedding-based)

2.2.5 Chatbot

- Provides book suggestions
- Assists with search queries
- Built using NLP frameworks

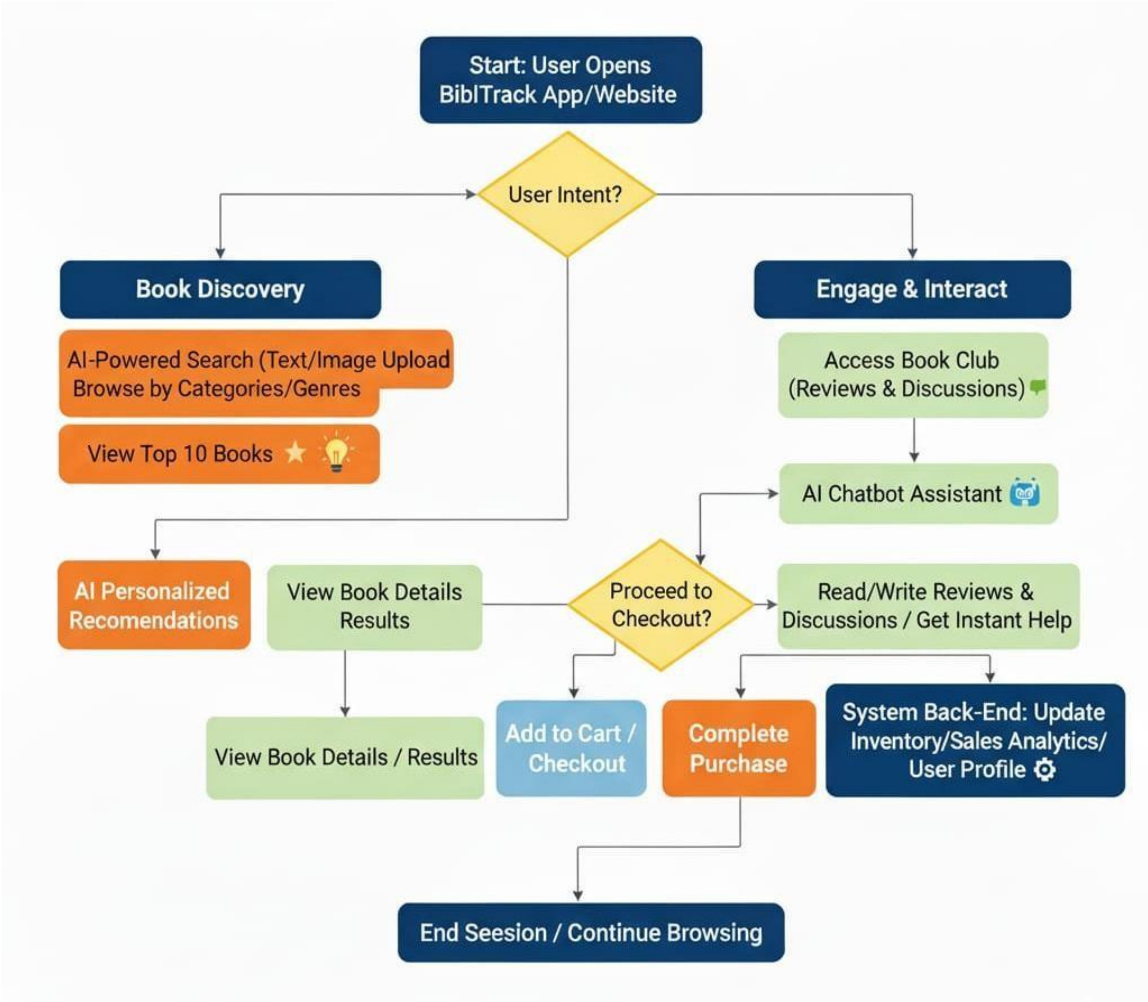

Fig 2.1 System Architecture of BIBLIOTRACK

2.2 Module Design

Bibliotrack is divided into functional modules:

2.2.1 User Authentication Module

- Login & registration
- Role-based access (user/admin)
- Maintains session authentication

2.2.2 Book Management Module

- Add/update/delete books
- Manage book metadata: author, genre, price

- Store cover images + generate visual embeddings

2.2.3 Visual Search Module

- User uploads a book cover/photo
- CNN + Image Hash model extracts features
- System finds the nearest matching book in the database

2.2.4 Recommendation Module

- Suggests books based on:
    - User's reading history
    - Genre similarity
    - Visual/semantic similarity embeddings
- Uses KNN + embedding similarity for prediction

2.2.5 Book Club & Community Module

- Users post reviews, discussions
- Join group reading clubs
- Admin moderates content

2.2.6 Chatbot Interaction Module

- Provides reading tips
- Helps with book discovery
- Supports multilingual queries

2.3 Database Design

Main Tables:

2.3.1 Users

- User details, passwords, roles

2.3.2 Books

- ISBN, title, author, genre
- Description
- Cover image path, embedding vectors
- Stock, price

2.3.3 Book Categories

- Genre & category mapping

2.3.4 Visual Embeddings

- Embedding vectors for visual search
- Image hash values

2.3.5 Recommendations

- Stores user-book match history
- Similarity scores

2.3.6 Book Club Posts & Comments

- Community posts
- Discussion threads
- Likes/replies

2.4 User Interface (UI) Design

Main Screens

2.4.1 Login & Registration Screen

- Secure authentication
- Role selection: User/Admin

2.4.2 Book Discovery Dashboard

- Search bar
- Recommended section
- Genre browsing

2.4.3 Visual Search Screen

- Upload book cover
- Instantly get book details

2.4.4 Book Club Screen

- Posts & comments

- Join reading discussions

2.4.5 Admin Dashboard

- Add/update/remove books
- Manage categories
- Monitor community content

2.4.6 Chatbot Screen

- AI assistant for real-time help

2.5 Technology Stack

2.5.1 Frontend

- HTML - 5
- CSS
- JavaScript

2.5.2 Backend

- Django (Python) – MVC architecture

2.5.3 AI/ML

- TensorFlow.js
- CNN models

2.5.4 NLP

- NLTK
- Regex pattern matching

2.5.5 Database

- SQLite

Chapter 3

IMPLEMENTATION

This chapter explains how Bibliotrack was implemented across backend, frontend, database, AI modules, and integrations.

3.1 Backend Implementation

The backend uses **Django REST Framework** to implement APIs.

Key API Endpoints

3.1.1 User APIs

- POST /login – Authenticate user
- POST /register – Register new user

3.1.2 Book & Inventory APIs

- POST /addBook – Add book & auto-generate embeddings
- GET /books – List all books
- GET /books/:id – Fetch single book details

3.1.3 Visual Search API

- POST /visualSearch – Upload image → returns best-matching book

3.1.4 Recommendation APIs

- GET /recommend/:userId – Personalized recommendations

3.1.5 Book Club APIs

- POST /post – Add community post
- GET /posts – View all discussions

3.2 Frontend Implementation

The frontend provides a clean and user-friendly interface for:

- Book browsing
- Visual search
- Book club discussions
- Chatbot queries

Main UI Components

- Login page
- Book discovery dashboard
- Visual search upload interface
- Recommendation carousel
- Admin inventory panel
- Book club feed
- Chatbot floating window

3.3 Database Implementation

Tables store:

3.3.1 Books

Stores all book information.

3.3.2 Users

Stores personal details, roles, preferences.

3.3.3 Visual Embeddings

Stores:

- CNN embedding vector
- Image hash
- Extracted feature metadata

3.3.4 Recommendations

Tracks:

- User history
- Match scores
- Suggested books

3.3.5 BookClub Posts

Stores:

- Post content
- Media
- User interactions

3.3.6 Chatbot Logs

- Stores user queries and chatbot replies
- Records timestamps of each interaction
- Saves recommended book IDs or links
- Helps improve chatbot accuracy
- Tracks common user requests
- Enables personalized book suggestions

Chapter 4

TESTING

This chapter covers the testing processes and methodologies applied to the BiblioTrack Bookstore project. Testing is essential to identify and correct any issues, validate that the system meets complex functional and non-functional requirements, and ensure that the AI/ML components (Search, Recommendations, Chatbot) perform reliably and accurately.

4.1 Testing Objectives

- Verify that the system functions as intended by executing a series of test cases.
- Ensure that all user roles (Admin and Customers) can access the intended features without any errors.
- Test the accuracy of search results, AI-based recommendations, and visual search output.
- Confirm that the system is secure and handles invalid inputs or unauthorized access appropriately.
- Evaluate the system's performance, reliability, and smooth integration between modules (frontend, backend, AI engines, and database).

4.2 Testing Environment

- **Hardware:** Laptop/PC with minimum 8GB RAM and multi-core processor.
- **Software:**
  - Backend: Django Framework
  - Frontend: Django Templates + JavaScripts
  - Database: SQLite
  - AI Modules: TensorFlow for visual search, Sentence Transformers for semantic search
  - Testing Tools: Postman for API testing, Selenium/Cypress for UI and end-to-end testing
- **Operating System:** Windows 10 / macOS / Linux
- **Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge for cross-browser testing

4.3 Types of Testing

4.3.1 Unit Testing

- **Objective:**
  To test individual functions or components in isolation to verify correctness.
- **Tools:**
  Python unittest, PyTest for backend logic, and simple front-end testing with JS test scripts.
- **Example Unit Test Cases**
  - **User Authentication:** Verifies that login and signup functions validate credentials correctly.
  - **Book Search Function:** Tests keyword search, filter operations, and sorting results.
  - **AI Semantic Search:** Checks that the semantic engine returns relevant results even when exact keywords do not match.
  - **Cart Operations:** Ensures add/remove/update item functions work accurately.
  - **Recommendation Engine:** Validates that recommendations match user history and preferences.

4.3.2 Integration Testing

- **Objective:**
  To test the interaction between different modules (Frontend → Backend → Database → AI Engines).
- **Example Integration Test Cases:**
  - **Search and AI Engine Integration:** Confirms that when a user searches and text results are insufficient, the system automatically triggers semantic or visual search modules.
  - **Order Placement Workflow:** Ensures the cart page, checkout process, backend order creation, and payment API integrate smoothly.
  - **User Access Control:** Confirms that only the admin can add/edit/delete books, and customers cannot access restricted pages.

- **Book Club Module:** Verifies that users can join clubs, post messages, and retrieve discussions from the database.

4.3.3 Functional Testing

- **Objective:**
  To validate the system against all functional requirements.
- **Test Scenarios:**
  - **Login and Registration:** Tests login and signup for all users and checks correct redirection to dashboards.
  - **Book Browsing & Filtering:** Verifies that users can view all books, apply filters, and sort results.
  - **Wishlist & Cart Operations:** Ensures that users can add books to wishlist, add to cart, and view/update their cart accurately.
  - **Visual Search:** Confirms that users can upload an image and the system displays visually similar books.
  - **Chatbot Interaction:** Tests whether the chatbot responds accurately to greetings, genre searches, and recommendation queries.
  - **Top Rated Books:** Checks that the system displays the top 10 books based on user ratings.
  - **Admin Book Management:** Verifies that the admin can add, edit, and delete books successfully.

4.4 Test Cases

Below are sample test cases for various component:

Table 4.4.1: Test Cases

Test Case ID	Description	Test Steps	Expected Result	status
TC-001	Login with valid credentials	Open login page. Enter a valid user name and password Click the **"Login"**	User is redirected to homepage with active session.	Pass
TC-002	Normal Search	Enter keyword in search bar.	Keyword matching books are displayed	Pass
TC-003	Semantic Search Trigger	Enter abstract query like book about space.	Semantic AI engine returns a similar result.	Pass
TC-004	AI recommendation Page	Goto recommendation page.	Personalized recommendation displayed based on searching history.	Pass
TC-005	Visual search	Upload cover image	System extract visual features and gives a similar result.	Pass
TC-006	Chatbot query	Enter a query like find mystery book	Chatbot lists a top mystery books.	Pass
TC-007	View book club	Navigate to book club	All posts load with correct user info.	Pass
TC-008	Top 10 books based on reviews	Navigate to Top10 book section	System correctly displays top 10 highest rated books.	Pass
TC-009	View book details	Click on any book	Book info loads with title, author, rating, price	Pass
TC-010	Admin manage books(Add/delete /edit)	Admin adds a new book And also edit and delete the book.	Book is successfully added,updated,and deleted.	Pass

The test cases in the table 4.4.1 thoroughly validate all core functionalities of the BiblioTrack system, including login, book browsing, semantic and visual search, wishlist, cart, and order processing. Admin capabilities such as adding, editing, and deleting books are verified to ensure proper backend management. AI-driven features like recommendations and chatbot responses are tested for accuracy and reliability. Community interactions through book clubs and the display of top- rated books ensure user engagement. Overall, the testing confirms that all integrated modules work smoothly and deliver a seamless user experience.

Chapter 5

RESULTS AND DISCUSSION

This chapter summarizes the results of the BiblioTrack intelligent book platform project, discussing its effectiveness, reliability, and alignment with the intended objectives. The chapter also covers challenges encountered, key insights gained, and recommendations for future improvements.

5.1 Results

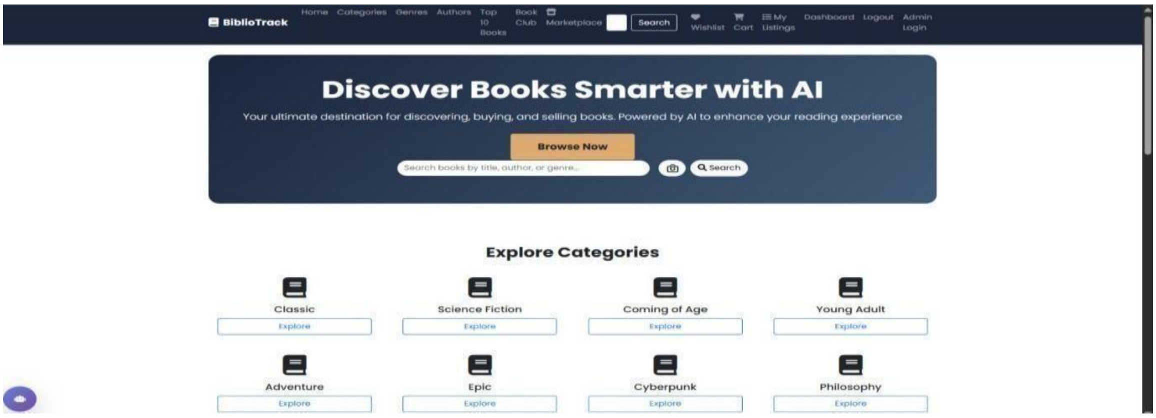

Fig 5.1.1 Home page of Bibliotrack

The fig 5.1.1 homepage introduces users to the platform with an AI-powered book discovery search bar, a prominent "Browse Now" button, and category-based exploration icons that help users find books quickly using text or visual search.

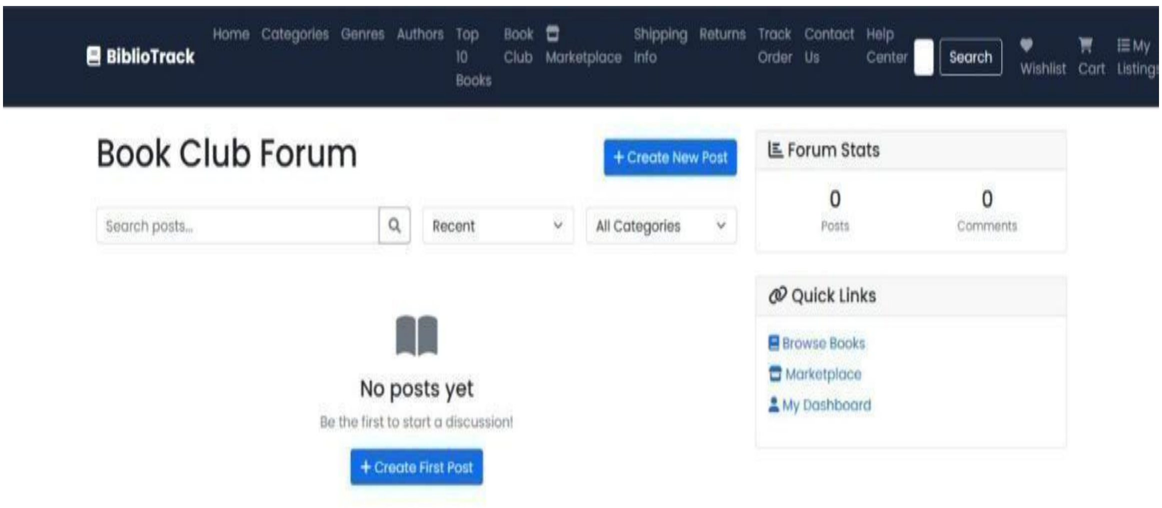

Fig 5.1.2 Book club of the website

The fig 5.1.2 displays the Book Club area where users can search posts, filter by category, and create new discussions.Forum statistics such as post count and comments are shown to indicate activity levels. Quick links allow users to easily navigate to browsing, the marketplace, or their dashboard.

Fig 5.1.3 Top 10 books on the basis of ratings

The fig 5.1.3 displays the top 10 trending books with ranking badges and book details such as author, price, and rating.Each book card includes actions like View Details and Add to Cart for quick engagement. The layout visually highlights popular books to encourage exploration and purchases

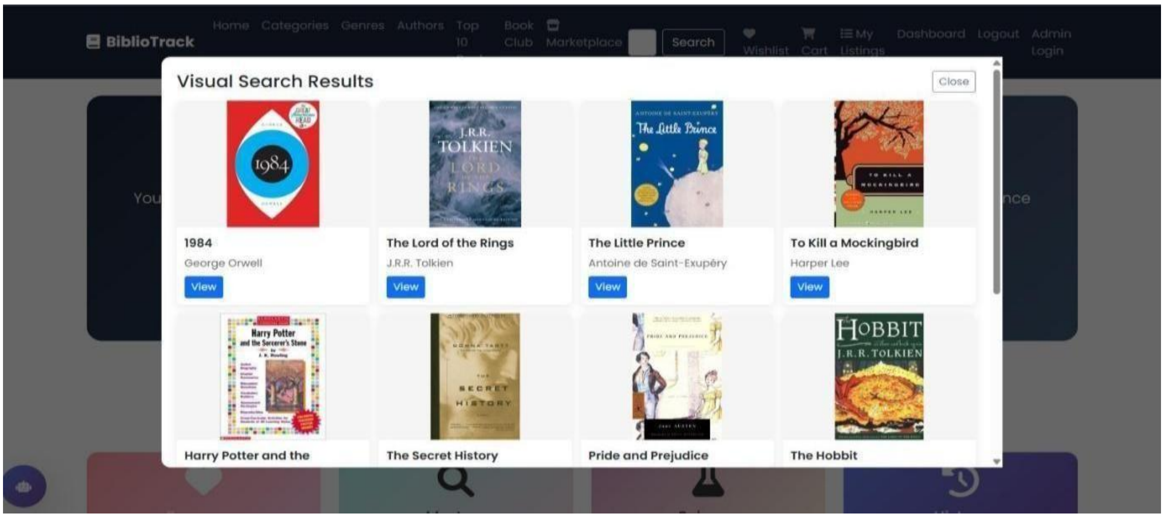

Fig 5.1.4 Instant visual search and recommendations through uploading book cover pages

The fig 5.1.4 shows books that visually match the uploaded image using AI-based feature extraction.Each result displays the book cover, title, author, and a View button for further details.The popup helps users discover books simply by scanning or uploading a cover image.

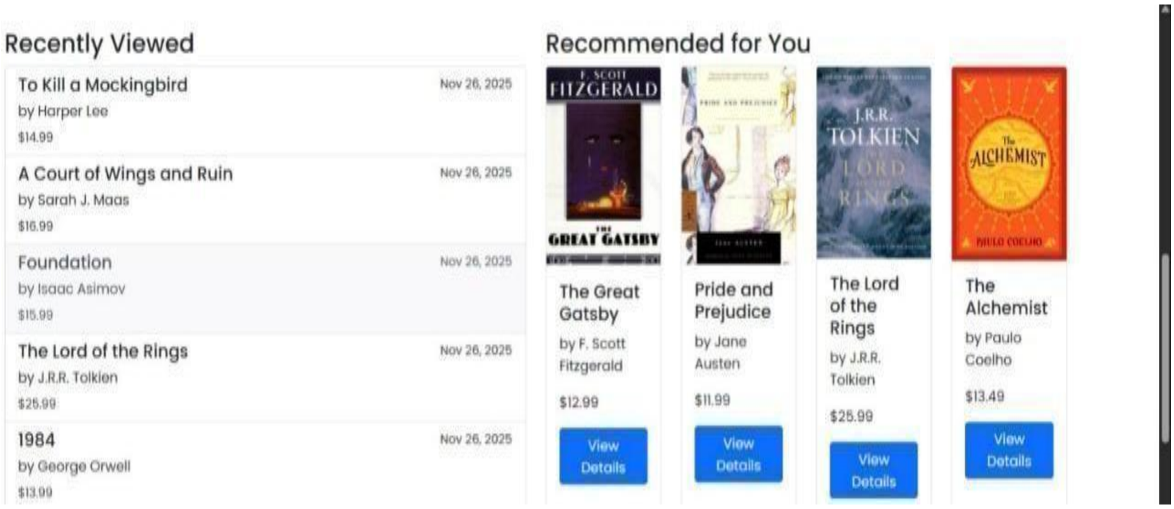

Fig 5.1.5 Recommendations on the basis of user interaction in dashboard

The fig 5.1.5 lists all books the user recently visited along with timestamps, giving them quick recall of their browsing history.A recommendation panel shows personalized book suggestions powered by AI similarity and user activity.Together, both sections help users continue their reading journey without losing context

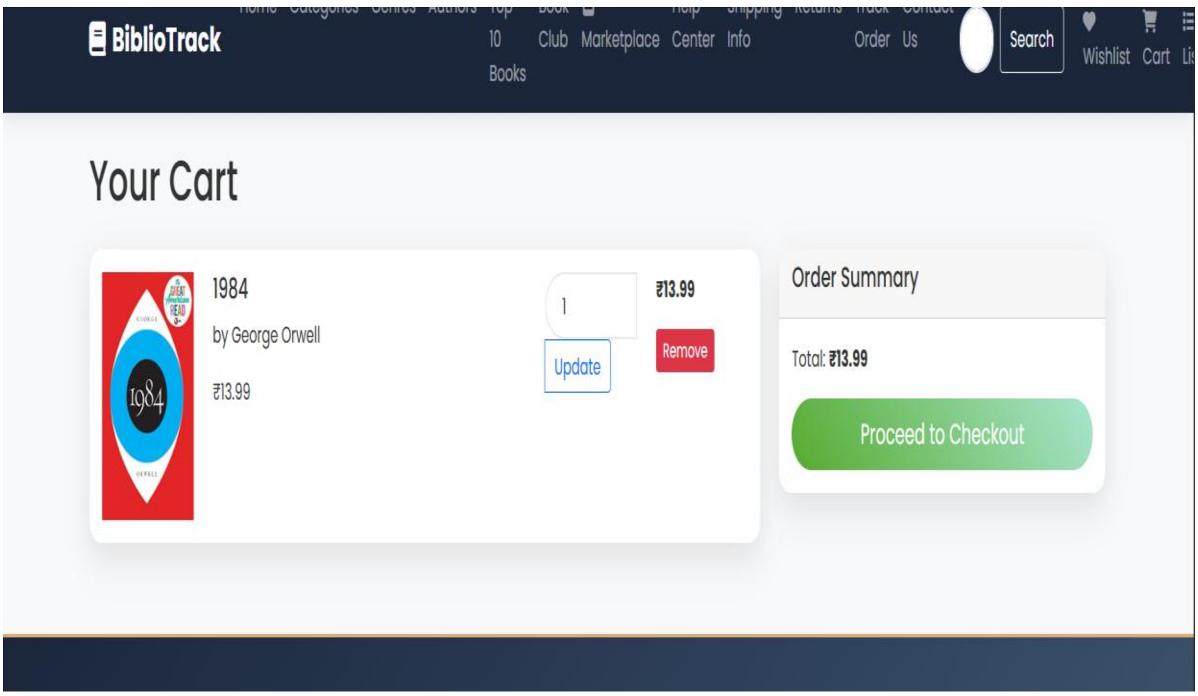

Fig 5.1.6 Cart section

The fig 5.1.6 displays all items added to the cart, showing book details, price, and options to update quantity or remove items.An order summary panel calculates the total price instantly.A clear Proceed to Checkout button guides users toward completing the purchase.

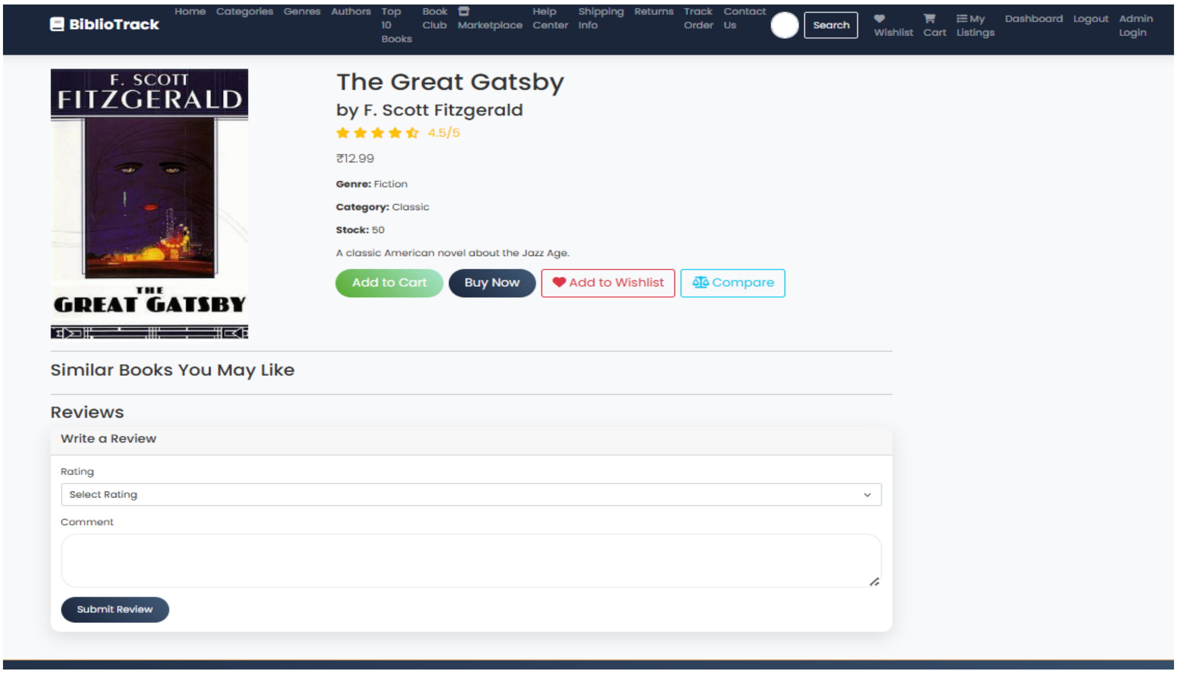

Fig 5.1.7 Book detail page

The fig 5.1.7 provides complete information about a selected book including title, author , genre, category, and stock availability.Buttons allow users to add the book to the cart, wishlist, or compare list.A Reviews section and similar book recommendations help user make informed purchase decisions.

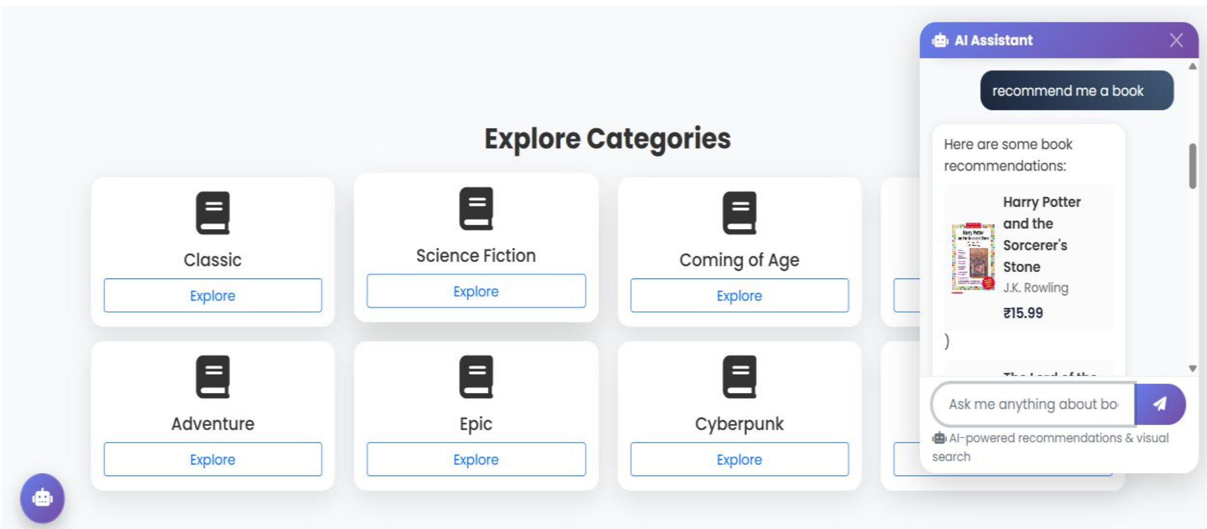

Fig 5.1.8 Book recommendations through chatbot

The chatbot shown in the fig 5.1.8 acts as an intelligent reading assistant that recommends books based on the user’s interests and genre preferences. It provides personalized suggestions instantly, helping readers discover new titles that match their tastes. By offering quick, AI-powered guidance, it makes exploring and choosing books easier and more enjoyable.

Fig 5.1.9 Book detail page

The fig 5.1.9 is an online book marketplace where users can browse available listings. It includes filtering options on the left and a featured book displayed in the main area. The interface is simple, focusing on helping users quickly find and view books for sale.

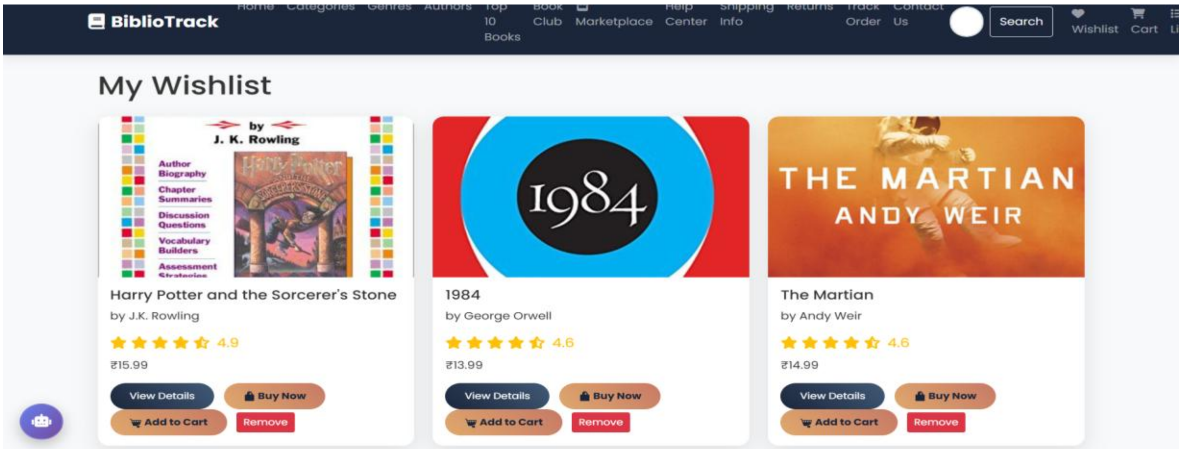

Fig 5.1.10 Wishlist section

The fig 5.1.10 is a wishlist section where users can save books they are interested in. Each book is presented in a card format with options to view more details, buy, or remove it from the list. The layout highlights user convenience and easy access to favorite books.

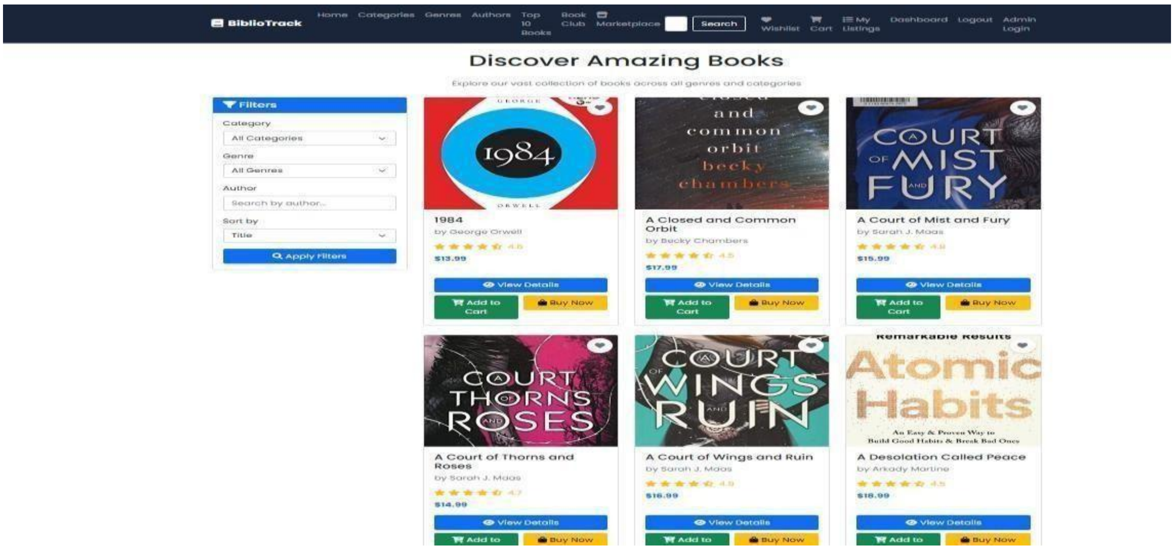

Fig 5.1.11 Discover books by filters like category,genre,price etc

The fig 5.1.11 shows a clean and modern online bookstore interface featuring a grid of popular book listings with cover images, prices, and user ratings. A filter panel on the left allows users to sort books by category, genre, author, and title. The overall layout is bright and user-friendly, emphasizing easy browsing and quick access to “Buy Now” and “Add to Cart” options.

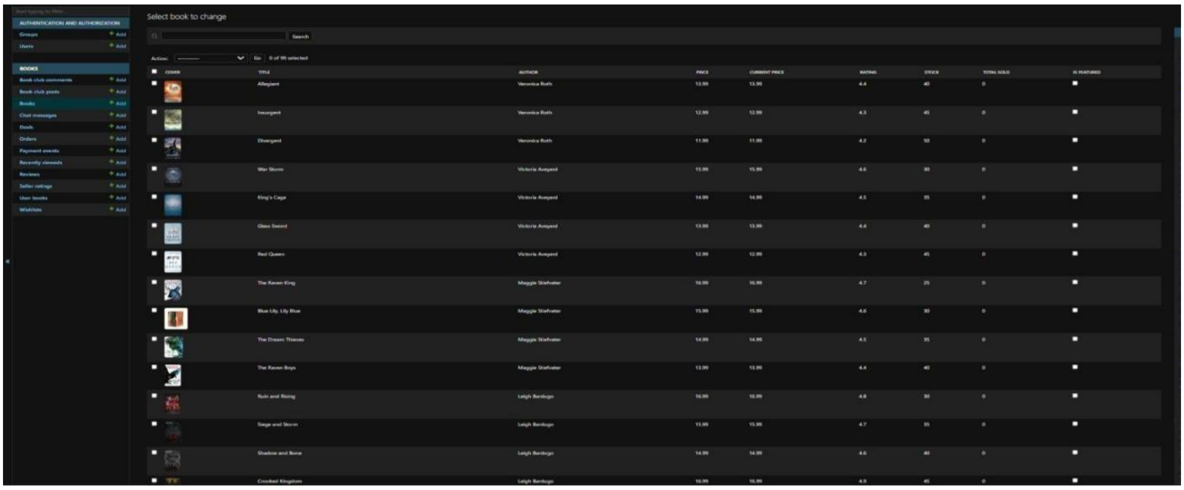

Fig 5.1.12 Admin Dashboard

The fig 5.1.12 displays an admin dashboard that lists books in a detailed table format, including information like author, price, rating, stock, and total sold. A dark-themed sidebar on the left provides navigation options for managing books, orders, reviews, users, and other administrative tasks. The interface is designed for efficiency, enabling administrators to update or manage large sets of book data quickly.

5.2 Discussion

Effectiveness of the System

- BiblioTrack effectively enhances the reading experience by making book discovery faster, tracking simpler, and engagement more interactive.
- Its AI chatbot and visual search provide personalized, accurate assistance, reducing user effort and improving satisfaction.
- The platform's community features and e-commerce integration create a seamless, all-in-one environment that boosts user involvement and supports both readers and sellers efficiently.

Challenges Encountered

- Users often struggle to find books matching their tastes because basic search and filters are too limited for nuanced preferences.
- Many discovery systems suffer from the "cold-start" problem — when a user is new, there is not enough data to generate reliable personalized recommendations.
- Recommendation engines tend to favor popular or mainstream books, which means readers interested in niche or long-tail titles rarely see relevant suggestions.
- Poor UI/UX design — such as confusing navigation, hidden search bars, or overloaded menus — often makes book discovery frustrating and slow.
- Insufficient metadata (e.g. limited genres, missing tags, or lack of book-cover based search) makes it hard for users to evaluate and discover books based on content, mood or context

Chapter 6

Conclusion and Future Enhancements

6.1 Conclusion

BiblioTrack is a smart and interactive book platform that helps users easily discover, track, and engage with books. It enhances the reading experience through AI-powered chatbot assistance and visual search, offering personalized support and quick book identification. Along with these, its community book club and e-commerce features create a complete, social, and convenient environment for book lovers.

6.2 Future Enhancements

- **Online Payment Gateway:**
  A secure system that allows customers to pay using various methods like cards, UPI, and wallets, ensuring smooth and safe transactions.
- **Mobile App Version:**
  A dedicated mobile application designed to give customers quick, easy, and convenient access to browse books, place orders, and track their activities anytime.
- **Analytics Dashboard:**
  A visual dashboard that provides meaningful insights on sales, user activity, and inventory, helping administrators make better decisions and manage the platform effectively.
- **Notifications:**
  An automated system that sends instant email or SMS alerts to users about their orders, status updates, new arrivals, and promotional offers.

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